

Juan José Jáuregui Barba

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Professional Summary

Recent graduate in Robotics and Digital Systems Engineering, focused on Industrial Automation. Builds software and embedded control systems that run real automated equipment, from real-time firmware to the control logic that coordinates it, incorporating AI where it adds real value. Distinctive strength in bridging software with embedded hardware control, validating systems end-to-end from simulation to physical deployment; published researcher (MICA 2025).

Education

Monterrey Institute of Technology and Higher Education (ITESM) 2022 – 2026
Robotics and Digital Systems Engineering GPA: 90/100
Relevant Coursework: Intelligent Robotics, Control Systems, Embedded Systems, Robotics Foundation

Experience

Research Intern August 2025 – January 2026
National Institute of Advanced Industrial Science and Technology (AIST) Tokyo, Japan

- Trained AI models for a robotic arm to grasp and handle objects under changing conditions.
- Built virtual environments to safely train and validate AI models before real-world use.
- Deployed trained AI models onto a real robotic arm with confirmed reliable performance.

Projects

Personal Project January 2025 – Present
AI-Driven Natural Language Robot Control Monterrey, Mexico

- Integrated an AI language model to convert voice commands into precise automated actions, reducing manual input.
- Designed and deployed the full solution end to end, achieving reliable execution across varied spoken commands.

ITESM | Manchester Robotics March 2026 – June 2026
Warehouse Automation & Autonomous Material Handling Monterrey, Mexico

- Led a 5-person team to develop the system that let an autonomous forklift robot track its own position in real time, updating 10 times per second.
- Designed an ArUco vision-based rescue layer that automatically corrected localization drift in low-feature corridors.
- Validated two full pick-and-place missions end-to-end on physical hardware, achieving 0.03 m goal-position accuracy with reliable obstacle avoidance.

ITESM | Manchester Robotics March 2025 – June 2025
Computer Vision for Process Automation Monterrey, Mexico

- Developed a computer vision system that identified objects to enable accurate, real-time decisions.
- Designed control logic that reliably coordinated multiple automated process steps without manual intervention.

ITESM | John Deere August 2024 – November 2024
Embedded Automation – Autonomous Waypoint Navigation Monterrey, Mexico

- Developed embedded electronics and software that enabled automated, hands-free waypoint navigation.
- Integrated multiple sensors and actuators into a real-time industrial control loop for coordinated operation.
- Combined sensor data to continuously and accurately track equipment position and movement.

Skills

Programming Languages: Python, C/C++, MATLAB, Rust, Bash.

Embedded & Controls: STM32, SPI, UART, CAN, PID Control, RTOS, Computer Vision.

Engineering Tools: SolidWorks, ROS 1/2, Docker, Linux, Git, MS Office, LLM API Integration.

Soft Skills: Team Leadership, Agile/Scrum, Problem-Solving, Self-Learning.

Languages: Spanish (Native), English (B2).

Certifications

Lean Six Sigma White Belt (2024) | SolidWorks CSWP (2025)